WELLBEING · SUSTAINING HEALTHY BEHAVIORS

Rebound

An adaptive engine that helps people start, keep up, and bounce back to healthier habits — so a bad week never resets progress to zero.

Product requirements document & 7-day build plan · Khaja Anwaruddin Hamza · 13 November 2026 · v1.0

Item	Detail
Challenge	Future Health Digital Innovation Challenge 2026 (Future Health – A Global Initiative by Abu Dhabi; on Hack2skill). Co-creators: Microsoft, Inception; collaborator: MBZUAI.
Problem statement	Wellbeing → Sustaining Healthy Behaviors: help people start, maintain and recover healthier behaviours with support that adapts to goals, circumstances, motivation and setbacks.
Deadline	**Working prototype + technical documentation + assumptions, dependencies and limitations — due 20 Nov 2026.** Evaluation 20–30 Nov. Top three present at GITEX, 7–11 Dec 2026.
Awards	USD 105,000 total: 50,000 / 35,000 / 20,000. Selected solutions get a readiness assessment and a potential pathway beyond the Challenge.
Data rules	Organisers provide no data (not even synthetic). Use only data we have the legal right to use, declare every source, avoid unneeded personal data.
Judged on	Health impact & clinical relevance · Technical depth & architecture · Working solution & demo quality · Responsible AI, safety & data governance · Scalability & deployment readiness · Commercial viability & path to pilot

Strategy in one line. Don't build another wellness app. Build the **adaptive behaviour-change loop** the brief says is missing, as an engine that plugs into the assets Abu Dhabi already has (app journeys, wearables, Health Age, workplace programmes). Prove it works with a **synthetic-population simulation** that measures sustained change and recovery after lapses, which is exactly what the brief asks for and what most teams won't show.

1. Problem

Awareness campaigns create short bursts of interest, but information alone rarely sustains change in movement, nutrition, sleep or smoking. Motivation fluctuates, routines get disrupted, and one lapse often leads to total disengagement. Generic programmes ignore language, culture, health literacy, work patterns (e.g. shift work), disability, family context and access. The building blocks already exist in the ecosystem. What's missing is a loop that:

- turns those assets into **small, feasible actions** sized to the person;
- **learns what works** for each individual over time;
- **catches drift early and supports recovery** after setbacks without blame;
- measures **sustained change**, not downloads, messages sent or short-lived engagement.

The key behavioural insight. Relapse-prevention research describes the *abstinence violation effect* (Marlatt & Gordon): after one slip, people conclude they've failed and give up entirely. Streak-based apps make this worse by resetting to zero. Meanwhile, habit-formation research (Lally et al., 2010) found that missing a single opportunity did not materially affect habit formation. Rebound is built around that gap: **a lapse is information, not failure.**

2. Solution overview

Rebound is an adaptive engine with a thin user app on top. Each day it gives the user one small action. It tracks a **habit-strength score** that decays gently instead of resetting. It detects **drift** before a full lapse, asks a 10-second blame-free check-in to identify the barrier, and adjusts the action's size, type, timing or channel. A contextual bandit learns per user which adjustments work, inside hard safety rules. Every suggestion carries a **“why am I seeing this?”** card naming the behaviour-change technique, its evidence and its limits.

User journey (demo narrative)

When	What happens to Ahmed (shift worker, Abu Dhabi; Arabic speaker; goal: move more, sleep better)
Day 1	Onboards in Arabic. Chooses goals, shares step count but not location, sets quiet hours around his shifts.
Weeks 1–2	Gets one micro-action a day sized to his baseline (“10-minute walk after your shift”). Taps “why this?” to see the technique and evidence.
Week 3	**Ramadan mode** switches on: action windows move to after iftar and sleep targets adjust. The old schedule is not counted as missed.
Week 5	Misses 4 days. State → *drifting* → *lapsed*. Gets “Rough week? What got in the way?” instead of a broken-streak message. Answers “tired”. Next action shrinks to a 3-minute stretch.
Week 6	Completes three small actions; state → *recovering* → *active*. Habit strength dipped but never reset. The dashboard logs a recovery time of 6 days.

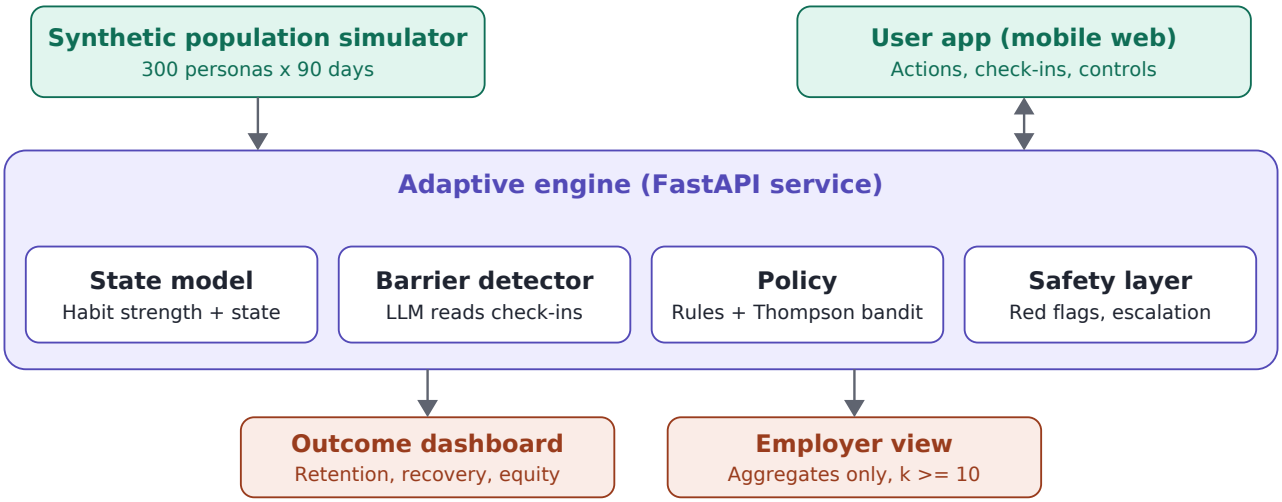
3. Goals and non-goals

Goals (this submission)	Non-goals (stated as future work)
• Working end-to-end prototype: onboarding → daily action → drift detection → check-in → adapted plan → recovery • Engine logic that is explainable and auditable • 90-day simulated evaluation vs. a streak-app baseline, with equity breakdown • Arabic + English, accessible UI, user data controls • Clear governance: consent, minimisation, employer aggregate-only view	• Clinical validation or real-user trial (needs ethics approval; proposed as pilot) • Native mobile apps (mobile web is enough for the demo) • Live integration with real wearables or government systems (designed as adapters, mocked) • Full nutrition or smoking modules (the engine is behaviour-agnostic; demo covers movement + sleep) • Medical diagnosis or treatment advice

4. Challenge requirements → features

#	Brief requirement	How Rebound meets it	Demo proof
1	Adaptive personal plans	Goal + baseline intake turns into tiered micro-actions (4 size tiers per behaviour). Context modes: Ramadan, summer heat, shift work, travel.	Onboarding flow; plan changes when a mode switches
2	Continuous learning & re-engagement	Engagement-state model (active / drifting / lapsed / recovering). Barrier check-in classified by LLM. Plan shrinks or swaps instead of resetting. Bandit learns per user.	Lapse → check-in → smaller action → recovery, shown live
3	Evidence-based behaviour support	Each action is tagged with a BCT Taxonomy v1 technique (Michie et al., 2013), e.g. implementation intentions, graded tasks, self-monitoring, prompts. A “why this?” card shows the rationale and the limits. Gamification is opt-in and non-punitive.	“Why this?” card; technique library in docs
4	User autonomy & inclusion	User owns goals, data sharing, notification windows and pause. Arabic + English (+ one South Asian language as a stretch). Voice-note option, large-text and screen-reader-friendly UI. Employer view is aggregate-only (k ≥ 10); no individual access.	Settings screen; employer dashboard showing only aggregates
5	Sustained outcome measurement	90-day simulation vs. baseline: persistence, recovery time after lapse, behaviour change, retention, equity by segment. The same metric definitions are reused in the proposed real pilot.	Outcome dashboard with curves and segment table

5. System architecture



Inputs reach the engine from the app (real interactions) or the simulator (synthetic users). Both run the **same engine code**, so the simulation evaluates the actual product logic, not a model of it. Outputs go to the user (next action, explanation) and to anonymised outcome stores that feed the dashboards.

6. Adaptive engine — functional spec

6.1 State model

- **Habit strength** per behaviour: $H_t = \lambda \cdot H_{t-1} + (1-\lambda) \cdot a_t$, where $a_t \in [0,1]$ is today's completion (partial credit for a smaller tier) and $\lambda \approx 0.9$. One missed day only dents H; it never resets.

- **Engagement state** from a 7-day window of completions, check-in responses and app opens: *active* (≥4/7), *drifting* (2–3/7 or a sharp drop vs. personal baseline), *lapsed* (0–1/7 or 3+ consecutive silent days), *recovering* (lapsed → 2+ completions in 4 days).
- Context flags: Ramadan window, heat index (simulated), shift pattern, travel/illness (self-reported), pause (user-set; never counted as failure).

6.2 Barrier detection

When the state becomes *drifting* or *lapsed*, the user gets a one-tap check-in ("What got in the way?") with buttons plus optional free text or voice. An LLM maps free text in any supported language to a fixed taxonomy: **time, energy/fatigue, illness/pain, schedule change, environment (heat, no safe space), low motivation, social/family, forgot**. It returns structured JSON with a confidence score. Low confidence falls back to the button answer.

6.3 Action library

Behaviours × size tiers × technique. Example for movement: Tier 1 "3-minute stretch" → Tier 2 "10-minute walk" → Tier 3 "20-minute brisk walk" → Tier 4 "30-minute activity of choice". Each action has metadata: tier, BCT code, rationale text, contraindications, context suitability (indoor/outdoor, fasting-safe, low-energy) and channel (app, WhatsApp-style message, voice note).

6.4 Policy

- **Layer 1: safety and context rules (always first).** Filter out unsafe or unsuitable actions (e.g. outdoor at midday in heat mode; strenuous while fasting; nothing after a pain/illness flag until the user says they're OK).
- **Layer 2: rule-based adaptation.** Barrier → adjustment: *time* → smaller tier + implementation intention; *energy* → lower tier + sleep-first; *schedule* → move the window; *motivation* → re-surface the user's own stated "why" + optional buddy; *forgot* → change prompt timing or channel.
- **Layer 3: contextual bandit (Thompson sampling).** Within the allowed set, learns per user which arm (tier × technique × time window × channel) yields completion. Reward = completion within 24h (primary) + engagement over the next 3 days (secondary). Priors are shared across similar personas so cold start is reasonable.
- **Never-reset principle:** after recovery, tier steps back up gradually (one tier per 3 successful days), never jumping straight back to the pre-lapse goal.

6.5 Safety layer

- Red-flag phrases (chest pain, fainting, breathlessness, suicidal language, signs of disordered eating) → stop coaching, show a supportive message and point to appropriate care or helplines. Logged for human review.
- No calorie or weight targets; no diagnosis; no medication advice. Smoking-cessation pharmacotherapy questions → refer to a clinician or pharmacist.
- LLM outputs are template-constrained (the LLM fills slots in approved message templates) and checked against a blocklist before sending.

7. User app — screens

Screen	Contents
Onboarding	Language (Arabic/English) → goals (movement, sleep; others shown as “coming soon”) → baseline questions → data-sharing toggles → notification window and quiet hours → optional context (shift worker, fasting).
Today	One action card with tier, time window, “why this?”, Done / Did a smaller version / Skip today. Habit-strength ring (decays smoothly; no streak counter).
Check-in	Shown when drifting or lapsed: one-tap barrier buttons + free text / voice. Non-judgemental copy, e.g. “Weeks like this happen. Let's make it easier.”
Why this?	Technique name (BCT code), plain-language rationale, evidence note, when it doesn't work well, and how to switch it off.
Progress	Habit strength over time, comebacks counted as wins (“3 comebacks this quarter”), behaviour trend.
Settings & data	What's collected and why, toggle each source, export/delete my data, pause programme, notification channel, accessibility (text size, high contrast, voice).

8. Simulator & evaluation

Because no real user data is available or permitted, we evaluate the engine on a synthetic population. All data is generated by our own code, so the rights question is settled and the declaration is simple. Every assumption is documented and adjustable.

Component	Specification
Personas	~300 synthetic users across segments: office worker, shift worker, outdoor/manual worker, homemaker/caregiver, student; Arabic / English / South Asian language; low/medium/high health literacy; with or without mobility limitation.
Hidden dynamics	Per-user latent motivation (random walk with mean reversion), habit strength, baseline capacity, and responsiveness to each technique (unknown to the engine; the bandit must learn it).
Disruptions	Ramadan (30 days, shifted windows), summer heat (outdoor penalty), travel, illness (random 3–7 day episodes), workload spikes. Rates vary by segment.
Comparators	(A) **Static streak app**: fixed goal, reminders, streak resets on miss. (B) **Rules-only Rebound** (no bandit). (C) **Full Rebound**. This ablation shows where the value comes from.
Run	90 simulated days, multiple random seeds; report mean ± spread.

Outcome metrics

Metric	Definition	Why it matters
Persistence	% of users with ≥3 active days/week in weeks 4, 8 and 12	Sustained change, not novelty
Recovery rate	% of lapses followed by return to active within 14 days	Core of brief requirement #2
Time to recover	Median days from lapse onset to recovering state	Speed of bounce-back
Behaviour change	Δ weekly active minutes; Δ sleep-timing regularity vs. baseline	Target behaviour, not engagement
Retention	% still interacting at day 90	Standard product metric, reported but not primary
Equity	All the above by segment; gap between best- and worst-served segment	“Equitable benefit” in the brief
Burden	Messages per user per week; % of messages in quiet hours (target 0)	Guards against manipulative over-engagement

9. Data, privacy & responsible AI

Area	Commitment
Data sources (declared)	(1) Synthetic data generated by our simulator; (2) optionally, one public, openly licensed wearable dataset used only to calibrate realistic step/sleep distributions (licence cited in docs); (3) LLM API used for language tasks on synthetic text only. No real personal data in the prototype.
Minimisation	No name, exact location, phone contacts or free-roaming sensors. Pseudonymous ID. Only behaviour signals the user turns on.
Consent & control	Per-source opt-in; plain-language “what we use and why”; pause, export and delete at any time; notification windows set by the user.
Third parties	Employers/insurers see only aggregates with minimum group size k ≥ 10; no individual-level data, rankings or penalties, by design and by API (no endpoint exists for it).
No dark patterns	No streak-loss fear messaging, no guilt copy, caps on message frequency, gamification opt-in only.
AI safety	Template-constrained LLM output, red-flag escalation, blocklist, human-review log; the bandit only ever chooses among pre-approved safe actions.
Transparency	“Why this?” on every action; model card + decision logic in the technical docs; limitations stated up front.

10. Tech stack

Layer	Choice	Notes
Engine & API	Python 3.11, FastAPI, NumPy/pandas, Pydantic	Single service; engine as a pure-Python package so the simulator imports it directly
Storage	SQLite (prototype) → Postgres	Event log of actions, completions, check-ins, state changes
Simulator	Python package + CLI; outputs Parquet/CSV	Seeded and reproducible; config in YAML
LLM layer	Azure OpenAI or any hosted model; Jais as an optional Arabic model	Barrier classification (JSON mode) + message slot-filling; provider-agnostic interface
Frontend	React (Vite) + Tailwind, mobile-first; Recharts for dashboards	RTL support for Arabic
Deploy	Frontend on Vercel; API on Render/Fly/Azure App Service	Public demo link for judges; seeded demo personas
Repo	GitHub monorepo: /engine /sim /api /web /docs	README with one-command local run

11. How this scores against the judging criteria

Criterion	Our evidence
Health impact & clinical relevance	Built on relapse prevention, habit formation and BCT Taxonomy; targets the exact failure mode named in the brief (disengaging after a lapse); persistence and behaviour-change metrics.
Technical depth & architecture	Layered policy (safety → rules → bandit), habit-strength state model, LLM barrier classifier, reproducible simulator with ablations.
Working solution & demo quality	Deployed app + live dashboard; a scripted 3-minute persona journey (Ahmed) plus simulated population results.
Responsible AI, safety & governance	Minimisation, consent controls, employer aggregates only, red-flag escalation, template-constrained LLM, explicit limitations.
Scalability & deployment readiness	Engine is an API that plugs into existing app journeys and wearable feeds; behaviour-agnostic action library; multilingual; WhatsApp-style channel adapter.
Commercial viability & path to pilot	B2G / B2B2C licensing per enrolled user for health-system and workplace programmes; 12-week pilot design ready (section 14).

12. Build plan: 13 → 20 November

Assumes one builder (solo) at roughly 6–8 focused hours a day. Riskiest work goes first so a demo-able core exists by Sunday. With teammates, split along the Owner column (E = engine/sim, F = frontend, D = docs/demo).

Day	Focus	Deliverables	Exit criteria	Owner
Fri 13	Foundations + engine core	Repo scaffold; action library (movement + sleep, 4 tiers, BCT tags); habit-strength + state model; rules policy	Unit tests pass; a single synthetic user can be stepped through 30 days in the CLI	E
Sat 14	Simulator + bandit	Persona generator, disruption events, baseline streak app, Thompson-sampling layer, metrics module	First 90-day run with curves for A vs. B vs. C; Full Rebound beats baseline on persistence and recovery	E
Sun 15	API + LLM layer	FastAPI endpoints (onboard, today, complete, check-in, state, explain); LLM barrier classifier + message templates (AR/EN); safety layer	End-to-end through the API with curl/Postman; red-flag tests pass	E
Mon 16	User app	Onboarding, Today, Check-in, Why-this, Progress, Settings; Arabic RTL; demo-persona time-travel control (“skip 7 days”)	Ahmed's journey clickable end to end on a phone	F
Tue 17	Dashboards	Outcome dashboard (persistence, recovery, behaviour Δ, equity table, burden); employer aggregate view (k ≥ 10)	Dashboard reads real simulator output; segment breakdown renders	F
Wed 18	Hardening + deploy	Context modes polish (Ramadan, heat, shift); accessibility pass; seeded demo data; deploy frontend + API	Public links work from a fresh browser; no console errors	E/F
Thu 19	Docs + demo	Technical documentation, assumptions/dependencies/limitations, data-source declaration, model card, pilot plan; 3-minute demo video	Docs complete against the submission checklist; video recorded	D
Fri 20	Buffer + submit	Fix list, final re-run of the simulation, README check, submit well before the cutoff (confirm time zone)	Submission confirmed on Hack2skill	All

Cut list if behind schedule (cut in this order): 3rd language → voice notes → employer view (keep as a mock screenshot) → bandit (keep rules-only, show the bandit as a simulated ablation) → Progress screen. **Never cut:** lapse → check-in → adapted plan flow, the simulator results, the safety layer, the documentation.

13. Submission checklist

Confirm against the full requirements sent at registration (check the Hack2skill dashboard or registration email).

	Item
☐	Working prototype: public demo URL + seeded demo personas + login-free access for judges

	Item
☐	Source repository with README (architecture, local run in one command, folder map)
☐	Technical documentation: architecture diagram, engine logic (state model, policy layers, bandit), API spec, simulator design
☐	Assumptions, dependencies and limitations statement (section 15, expanded)
☐	Data-source declaration: every dataset/API, licence, and what it was used for
☐	Responsible AI & safety note: guardrails, red flags, governance, employer restrictions
☐	Evaluation results: simulation method, metrics, A/B/C comparison, equity breakdown, caveats
☐	Pilot and commercial plan (1 page)
☐	Demo video (~3 min) if required or allowed

14. Path to pilot & commercial model

- **Pilot design (12 weeks):** one workplace cohort or one primary-care patient-app cohort (n ≈ 200–500). Micro-randomised design (as in the HeartSteps study, Klasnja et al., 2019) to test which adaptations work, plus a comparison against the existing programme. Ethics approval and clinical oversight before launch.
- **Pilot success metrics:** the same as section 8 (persistence at weeks 4/8/12, recovery rate, time to recover, behaviour change, equity gap, burden), plus user-trust and satisfaction surveys.
- **Integration:** Rebound as an API/SDK embedded in existing app journeys and wearable feeds; the operator keeps the user relationship; data stays in-region.
- **Business model:** licence per enrolled user per month for health-system and workplace-wellness programmes; outcome-linked pricing tier once persistence is validated.

15. Assumptions, dependencies & limitations

Type	Statement
Assumption	Simulated behaviour dynamics (motivation drift, disruption rates, technique responsiveness) are plausible but not empirically fitted to Abu Dhabi populations; results show relative advantage, not real-world effect sizes.
Assumption	Users will answer short check-ins at a meaningful rate; the system falls back to passive signals if they don't.
Assumption	Wearable step and sleep signals are available via existing platforms (e.g. phone health APIs) through adapters in production; mocked in the prototype.
Dependency	A hosted LLM API for barrier classification and message generation (provider-swappable); hosting for the frontend and API.
Dependency	Clinical review of the action library, red-flag list and messaging before any real-user pilot.
Limitation	No real users or clinical validation; the bandit's real learning speed is unknown until piloted.
Limitation	Prototype covers movement and sleep only; nutrition and smoking-cessation modules are designed but not built.
Limitation	LLM classification quality in Arabic dialects and South Asian languages is not formally benchmarked in this prototype.
Limitation	Not a medical device; provides no diagnosis or treatment advice.

16. Risks & mitigations

Risk	Mitigation
Time: 7 days is tight for engine + app + docs	Engine and simulator first; strict cut list; docs written incrementally from Day 1 notes
Simulation seen as “rigged”	Publish assumptions and config; include ablations and a case where Rebound doesn't help; seeds reproducible
LLM misclassifies or says something unsafe	Button answer as primary; JSON schema + confidence threshold; template-constrained output; blocklist
Judges see “just another wellness app”	Lead the demo with the lapse-recovery moment and the outcome curves, not the onboarding

17. Open questions

- Team: solo or up to 5 members? Split the build plan by owner once confirmed.
- Exact submission format and cutoff time/time zone (from the registration materials).
- Which LLM provider we have access to (Azure OpenAI / other); whether Jais is accessible for Arabic.
- Whether to add a third language (Hindi/Urdu or Malayalam) or spend that time on demo polish.

References cited: Marlatt & Gordon, Relapse Prevention (1985); Lally et al., “How are habits formed”, Eur. J. Soc. Psychol. (2010); Michie et al., “Behavior Change Technique Taxonomy v1”, Ann. Behav. Med. (2013); Klasnja et al., “Efficacy of contextually tailored suggestions for physical activity: a micro-randomized optimization trial of HeartSteps”, Ann. Behav. Med. (2019). Challenge details from the Future Health Digital Innovation Challenge page on Hack2skill.